Silicon Validation Engineer

Electronics & Programming
Complete Concept, Formula and Problem-Solving Guide

12 topics • key formulae • worked examples • practice problems • answer key • interview quick-fire

How to use this guide
1. Read each topic's concept summary, then memorise the yellow **formula box**. 2. Cover the solution of every blue **worked example** and try it yourself first. 3. Do the **practice problems** on paper, then check the Answer Key at the end. 4. Finish with the Silicon-Validation context chapter and the one-page cheat sheet. *Convention: $V_T = kT/q \approx 25.85$ mV at 300 K (use 26 mV for quick estimates). Mains frequency = 50 Hz.*

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1. Linear Components – R, C, L

Syllabus: Resistor, capacitor, inductor – units, V–I characteristics (linear, lagging, leading), response with time and frequency, behaviour towards DC and AC.

1.1 Concepts

	Resistor (R)	Capacitor (C)	Inductor (L)
Unit (symbol)	Ohm (Ω)	Farad (F) – usually pF, nF, µF	Henry (H) – usually nH, µH, mH
V–I law	v = i·R (linear, in phase)	i = C·dv/dt	v = L·di/dt
Phase (AC)	V and I in phase (0°)	**Current LEADS** voltage by 90°	**Current LAGS** voltage by 90°
Impedance	Z = R	Z = 1/(jωC), X_C = 1/(2πfC)	Z = jωL, X_L = 2πfL
At DC (steady state)	Same R	**Open circuit** (blocks DC)	**Short circuit** (passes DC)
At very high f	Same R	Short circuit	Open circuit
Stored energy	None (dissipates I^2R)	½CV^2	½LI^2
Cannot change instantly	—	Voltage across C	Current through L

Memory trick: **“ELI the ICE man”** – in an inductor (L) voltage E leads current I; in a capacitor (C) current I leads voltage E. Capacitors and inductors have *frequency-dependent* impedance, hence “lagging/leading” behaviour; resistors do not.

1.2 Time response (transients)

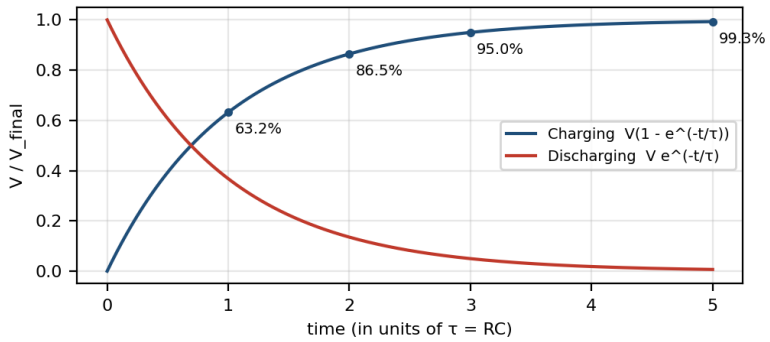


Fig 1.1 – RC charging and discharging. After 1τ: 63.2 %, 2τ: 86.5 %, 3τ: 95 %, 5τ: 99.3 %.

Key formulae	
Time constants	RC circuit: τ = R·C | RL circuit: τ = L/R
RC charging	$v_C(t) = V_s(1 - e^{-t/\tau})$
RC discharging	$v_C(t) = V_0 \cdot e^{-t/\tau}$
RL current rise	$i(t) = (V/R)(1 - e^{-t/\tau})$
Time to reach x %	$t = -\tau \cdot \ln(1 - x/100)$; 10→90 % rise time $t_r = 2.2\tau$; $t_r \approx 0.35/BW$
Reactances	$X_C = 1/(2\pi fC)$; $X_L = 2\pi fL$; $\lvert Z\rvert = \sqrt{(R^2 + (X_L - X_C)^2)}$
Series / parallel	R: $R_s = \Sigma R$, $1/R_p = \Sigma 1/R$. C: opposite (parallel adds, series is $1/\Sigma C^{-1}$). L: like R
Resonance (RLC)	$f_0 = 1/(2\pi\sqrt{(LC)})$; $Q_{series} = (1/R)\sqrt{(L/C)} = X_L/R$; $BW = f_0/Q$
Power	$P = VI = I^2R = V^2/R$; AC: $P = V_{rms}I_{rms}\cos\varphi$; $V_{rms} = V_p/\sqrt{2}$
RC corner frequency	$f_c = 1/(2\pi RC)$ (also for RL: $f_c = R/(2\pi L)$)

Worked example – RC time response

Problem: R = 10 kΩ, C = 100 nF is driven by a 5 V step. Find τ, the time to reach 90 % and the corner frequency.

Solution: τ = RC = 10k × 100n = **1 ms**. $t_{90} = \tau \cdot \ln 10 = 2.303 \times 1$ ms = **2.3 ms**. $f_c = 1/(2\pi \times 1\text{ ms})$ = **159.2 Hz**.

Worked example – Reactance
Problem: Find X_C of 1 µF and X_L of 10 mH at 1 kHz.
Solution: $X_C = 1/(2\pi \cdot 1000 \cdot 1\times10^{-6}) =$ **159.2 Ω**. $X_L = 2\pi \cdot 1000 \cdot 0.01 =$ **62.8 Ω**.

Worked example – Resonance
Problem: L = 10 mH, C = 100 nF. Find f_0.
Solution: $LC = 10^{-2}\times10^{-7} = 10^{-9}$; $\sqrt{} = 3.162\times10^{-5}$; $f_0 = 1/(2\pi \times 3.162\times10^{-5}) =$ **5.03 kHz**.

Practice problems (answers in the Answer Key)

Q1.1 A 1 kΩ resistor and 1 µF capacitor are driven by a 5 V step. Find v_C at t = 2 ms.
Q1.2 The current through a 100 mH inductor ramps from 0 to 2 A in 40 ms. Find the (constant) voltage across it.
Q1.3 Energy stored in a 10 µF capacitor charged to 12 V?
Q1.4 Series RLC: R = 10 Ω, L = 1 mH, C = 1 µF. Find f_0 and Q.
Q1.5 X_C of a 100 nF capacitor at 10 kHz?
Q1.6 Time constant of L = 10 mH, R = 100 Ω?
Q1.7 10 µF and 40 µF capacitors are in series. Equivalent capacitance?
Q1.8 A sine voltage is applied across an ideal capacitor. What is the phase of the current?

2. Network and Circuit Theory

Syllabus: Problem solving with passive components and DC sources; voltage and current dividers; series/parallel combinations.

2.1 Laws and theorems

- **Ohm's law:** V = IR. **KCL:** sum of currents entering a node = 0. **KVL:** sum of voltages around a closed loop = 0.
- **Superposition** (linear circuits): turn on one source at a time – other voltage sources → short, other current sources → open – then add the results.
- **Thevenin:** any linear two-terminal network = V_{th} (open-circuit voltage) in series with R_{th} (resistance seen with sources turned off). **Norton:** $I_N = V_{th}/R_{th}$ in parallel with R_{th}.
- **Maximum power transfer:** load gets maximum power when $R_L = R_{th}$; efficiency is then only 50 %.
- **Nodal analysis** (KCL at nodes) is best with many parallel branches; **mesh analysis** (KVL on loops) for many series loops.
- **Delta–Wye:** for equal resistors $R_Y = R_\Delta/3$.

Key formulae	
Series resistors	$R_{eq} = R_1 + R_2 + \ldots$
Parallel resistors	$1/R_{eq} = 1/R_1 + 1/R_2 + \ldots$; two resistors: $R_1R_2/(R_1+R_2)$
Voltage divider	$V_{out} = V_{in} \cdot R_2/(R_1 + R_2)$ (R_2 = resistor across which V_{out} is taken)
Current divider (2 branches)	$I_1 = I_{total} \cdot R_2/(R_1 + R_2)$ – the *other* resistor is on top
Loaded divider	Replace R_2 by $R_2 \parallel R_L$
Power	$P = VI = I^2R = V^2/R$
Max power transfer	$R_L = R_{th}$; $P_{max} = V_{th}^2/(4R_{th})$
LED resistor	$R = (V_{supply} - V_F)/I_{LED}$

Worked example – Voltage divider
Problem: V_{in} = 12 V, R_1 = 10 kΩ, R_2 = 5 kΩ. Find V_{out} across R_2, and V_{out} if a 5 kΩ load is attached.
Solution: Unloaded: 12 × 5/15 = **4 V**. Loaded: $R_2 \parallel R_L$ = 2.5 kΩ → 12 × 2.5/12.5 = **2.4 V** (loading effect – a divider needs a load ≫ R_2).

Worked example – Current divider
Problem: 9 mA flows into a 1 kΩ ∥ 2 kΩ pair. Find each branch current.
Solution: I_{1k} = 9 × 2/(1+2) = **6 mA**; I_{2k} = 9 × 1/3 = **3 mA** (more current in the smaller R).

Worked example – Thevenin + max power
Problem: A 12 V source with 6 Ω series resistor feeds a node that has a 3 Ω resistor to ground. The load connects across the 3 Ω. Find V_{th}, R_{th}, and the maximum load power.
Solution: V_{th} = 12 × 3/(6+3) = **4 V**. R_{th} = 6 ∥ 3 = **2 Ω**. For R_L = 2 Ω: $P_{max} = 4^2/(4\times2)$ = **2 W**.

Worked example – Parallel resistors
Problem: Find 100 Ω ∥ 220 Ω ∥ 330 Ω.
Solution: G = 0.01 + 0.004545 + 0.003030 = 0.017576 S → R = **56.9 Ω** (always less than the smallest branch).

Practice problems (answers in the Answer Key)

Q2.1 V_{in} = 5 V, R_1 = 2.2 kΩ (top), R_2 = 3.3 kΩ (bottom). Find V_{out}.
Q2.2 Choose a resistor for a 2 V, 10 mA LED on a 5 V rail.
Q2.3 Find R_{eq} of (4 Ω ∥ 12 Ω) in series with 5 Ω.
Q2.4 10 V source, 4 kΩ in series, 6 kΩ to ground. Find the Thevenin equivalent seen at the 6 kΩ and max power into a matched load.
Q2.5 At a node 5 mA enters, 2 mA and 1 mA leave through two branches, and a third branch carries I out. Find I.
Q2.6 Is a ¼ W 220 Ω resistor safe with 5 V across it?
Q2.7 Delta of three 30 Ω resistors → equivalent Wye value?

3. Diodes, Transistors and Op-amps – Types and Applications

Syllabus: types, applications and working principle – *when to use what* and how it works (clamp, clipper, protection ...).

3.1 Diode types – when to use what

Type	Key property	Typical use
Silicon rectifier (1N400x)	$V_F \approx 0.7$ V, slow recovery	Mains rectification, reverse-polarity protection
Schottky	$V_F \approx 0.2$–0.4 V, very fast, higher leakage	Low-drop rectifier in DC-DC converters, OR-ing, high-speed clamps
Zener	Conducts in reverse at V_Z (sharp breakdown)	Shunt voltage reference/regulator, over-voltage clamp
LED	V_F 1.8–3.3 V, emits light	Indication; always use a series resistor
Photodiode	Reverse-biased, current ∝ light	Sensors, optical receivers
TVS diode	Fast avalanche clamp	ESD / surge protection on I/O pins
Varactor	Capacitance varies with reverse V	VCOs, tuners
Fast-recovery / switching (1N4148)	t_{rr} of ns	Signal clipping, logic, snubbers

Key formulae	
Shockley equation	$I = I_S(e^{V/(nV_T)} - 1)$; $V_T = kT/q \approx 25.85$ mV @ 300 K; ≈ 60 mV per decade of current (n=1)
Half-wave rectifier	$V_{dc} = V_m/\pi$; $V_{rms} = V_m/2$; PIV = V_m; ripple factor 1.21; $\eta_{max} = 40.6$ %
Full-wave rectifier	$V_{dc} = 2V_m/\pi$; $V_{rms} = V_m/\sqrt{2}$; ripple factor 0.48; $\eta_{max} = 81.2$ %; PIV: bridge = V_m, centre-tap = $2V_m$
Bridge output peak	$V_{m,out} = V_{m,secondary} - 2V_F$
Filter-cap ripple	$V_{r(pp)} \approx I_{load}/(f_{ripple} \cdot C)$; $f_{ripple} = 100$ Hz for full-wave on 50 Hz mains
Zener regulator	$R_s = (V_{in} - V_Z)/(I_{Z,min} + I_{L,max})$; $P_Z = V_Z \cdot I_Z$
Diode + series R	$I = (V_s - V_F)/R$

3.2 Clippers, clampers and protection

- **Clipper (limiter):** removes part of the waveform above/below a level; shape is cut. Series diode clipper: diode blocks half the cycle. Shunt clipper: diode + DC reference conducts above (V_{ref} + 0.7 V) and flattens the peak. *Use:* protecting ADC/input pins from over-voltage.
- **Clamper (DC restorer):** shifts the whole waveform up/down without changing its shape – capacitor charges to the peak through a diode. A negative clamper (clamps positive peaks to 0 V) gives $-2V_m$ ... 0 V for a sine of peak V_m. *Use:* restoring DC level in video, level shifting.
- **Protection uses:** reverse-polarity diode in series (or P-MOSFET), **flyback/freewheel diode** across a relay or motor coil (absorbs the inductive spike V = L·di/dt), TVS/Zener on I/O for ESD, Schottky diode OR-ing for redundant supplies, input clamps to V_{DD}/GND.

Worked example – Rectifier with filter

Problem: A 50 Hz bridge rectifier has a 12 V_{rms} secondary, V_F = 0.7 V per diode, I_{load} = 500 mA, C = 2200 µF. Find DC peak and ripple.

Solution: $V_m = 12 \times \sqrt{2} = 16.97$ V; peak out = 16.97 − 1.4 = **15.57 V**. $f_{ripple} = 100$ Hz. $V_r = 0.5/(100 \times 2.2 \times 10^{-3}) =$ **2.27 V p-p**.

Worked example – Zener regulator design

Problem: V_{in} = 12 V, V_Z = 5.1 V, I_L = 20 mA, $I_{Z,min}$ = 5 mA. Find R_s and worst-case Zener power.

Solution: $R_s = (12 - 5.1)/(25\text{ mA}) = 276\ \Omega$ → use **270 Ω**. With load removed all current goes through the Zener: I = 6.9/270 = 25.6 mA, $P_Z = 5.1 \times 25.6$ mA ≈ **130 mW**.

Practice problems (answers in the Answer Key)

Q3.1 By what factor does a diode current change when V increases by 60 mV (n = 1, 300 K)?

Q3.2 Ideal half-wave rectifier, 10 V peak sine. Find V_{dc} and V_{rms} of output.

Q3.3 Ideal full-wave rectifier, 10 V peak. Find V_{dc} and V_{rms}.

Q3.4 Zener (5.1 V) fed from 9 V through 220 Ω with no load. I_Z and P_Z?

Q3.5 A 5 V_{peak} sine passes through a clamper that clamps positive peaks to 0 V (ideal). Output range?

Q3.6 5 V supply, silicon diode and 1 kΩ in series. Current?

Q3.7 What is the PIV of each diode in a bridge rectifier with secondary peak V_m?

Q3.8 Why is a flyback diode needed across a relay coil?

4. Transistors – BJT, FET and CMOS

Syllabus: pnp/npn characteristics and working principle; CE, CB, CC modes and advantages; FET, CMOS; simple transistor circuits – problem solving (given gain / β / h_{fe}).

4.1 BJT fundamentals

- **Regions:** cut-off ($V_{BE} < 0.6$ V, $I_C \approx 0$), **active** (BE forward, BC reverse – amplifier), **saturation** (both forward, $V_{CE(sat)} \approx 0.2$ V – closed switch).
- **npn** conducts when base is ~0.7 V above emitter (current flows C → E); **pnp** conducts when base is 0.7 V below emitter (E → C); all polarities reversed.
- β (= h_{FE} DC, h_{fe} small-signal) typically 50–300 and varies with temperature and I_C – good designs do not depend on it (use emitter degeneration).

Key formulae	
Currents	$I_E = I_B + I_C$; $I_C = \beta I_B$; $I_E = (\beta+1)I_B$
α and β	$\alpha = I_C/I_E = \beta/(\beta+1)$; $\beta = \alpha/(1-\alpha)$
Small-signal	$g_m = I_C/V_T$; $r_\pi = \beta/g_m$; $r_e = V_T/I_E \approx 1/g_m$
Voltage-divider bias	$V_{th} = V_{CC}R_2/(R_1+R_2)$; $R_{th} = R_1 \| R_2$; $I_B = (V_{th} - 0.7)/(R_{th} + (\beta+1)R_E)$
Approximate bias	$V_B \approx V_{th}$; $V_E = V_B - 0.7$; $I_E = V_E/R_E \approx I_C$; $V_{CE} = V_{CC} - I_C(R_C+R_E)$
CE amplifier gain	$A_v = -g_m(R_C \| R_L)$ (emitter bypassed); with unbypassed R_E: $A_v \approx -R_C/(r_e + R_E)$
Switch design	$I_B \geq I_{C,sat}/\beta_{min}$ (use 2–5× overdrive); $R_B = (V_{drive} - 0.7)/I_B$

	Common Emitter (CE)	**Common Base (CB)**	**Common Collector (CC) / emitter follower**
Input / output	Base / Collector	Emitter / Collector	Base / Emitter
Voltage gain	High, **inverting** (180°)	High, non-inverting	≈ 1 (slightly < 1), non-inverting
Current gain	β (high)	α ≈ 1	β + 1 (high)
Z_{in} / Z_{out}	Medium / medium-high	**Low** / high	**High** / **low**
Advantage / use	General-purpose amplifier – highest power gain	Best high-frequency response (no Miller effect); current buffer, RF amps	Buffer / impedance matching, driving low-impedance loads

4.2 MOSFET / FET and CMOS

- MOSFET is **voltage-controlled** (no DC gate current, very high input impedance); BJT is current-controlled.
- **n-MOS** turns ON for $V_{GS} > V_{th}$; **p-MOS** turns ON for $V_{GS} < -|V_{th}|$. Regions: cut-off, triode (resistor-like, switch ON), saturation (amplifier / current source).
- **JFET** is normally ON (depletion mode); gate-source junction must stay reverse biased; $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$.
- **CMOS inverter:** p-MOS pull-up + n-MOS pull-down; only one conducts in steady state → near-zero static power; dissipates only while switching (dynamic power) and by leakage. Switching threshold $V_M \approx V_{DD}/2$ for balanced sizing.
- BJT vs MOSFET: BJT – higher g_m per current, good analog precision; MOSFET – low power, easy to integrate (dominant in digital ICs), simple switch.

Key formulae	
nMOS saturation ($V_{DS} \geq V_{ov}$)	$I_D = ½\cdot\mu_n C_{ox}(W/L)(V_{GS} - V_{th})^2$ (= ½k$(V_{GS} - V_{th})^2$)
nMOS triode ($V_{DS} < V_{ov}$)	$I_D = k[(V_{GS} - V_{th})V_{DS} - V_{DS}^2/2]$
Transconductance	$g_m = k\cdot V_{ov} = 2I_D/V_{ov}$, with $V_{ov} = V_{GS} - V_{th}$
MOSFET CS gain	$A_v = -g_m(R_D \| R_L \| r_o)$
CMOS dynamic power	$P_{dyn} = \alpha\cdot C_L\cdot V_{DD}^2\cdot f$; Short-circuit + leakage extra

Worked example – Voltage-divider bias (given β)

Problem: V_{CC} = 12 V, R_1 = 47 kΩ, R_2 = 10 kΩ, R_C = 2.2 kΩ, R_E = 1 kΩ, β = 100. Find I_C, V_{CE}, g_m, and mid-band gain with R_E bypassed.

Solution: V_{th} = 12×10/57 = 2.105 V; R_{th} = 47k‖10k = 8.25 kΩ. I_B = (2.105 − 0.7)/(8.25k + 101×1k) = 12.9 µA → **I_C = 1.29 mA**, I_E = 1.30 mA. V_E = 1.30 V, V_C = 12 − 1.29m×2.2k = 9.17 V → **V_{CE} = 7.87 V** (active). g_m = 1.29 mA/25.85 mV = 49.7 mS; A_v = −49.7m × 2.2k ≈ **−109**.

Worked example – BJT as a switch

Problem: Drive a 100 mA load from a 3.3 V GPIO using a BJT with β_{min} = 50.

Solution: $I_{B,min}$ = 100/50 = 2 mA; design for 5 mA (overdrive). R_B = (3.3 − 0.7)/5 mA = 520 Ω → **510 Ω**. Add a flyback diode if the load is inductive. (Check GPIO can source 5 mA; otherwise use a MOSFET.)

Worked example – nMOS current

Problem: $k = \mu_n C_{ox} W/L$ = 2 mA/V², V_{th} = 1 V, V_{GS} = 3 V, V_{DS} = 5 V. Find I_D and g_m.

Solution: V_{ov} = 2 V < V_{DS} → saturation. I_D = ½×2m×2² = **4 mA**; g_m = 2m×2 = **4 mS**.

Practice problems (answers in the Answer Key)

Q4.1 I_B = 20 µA, I_C = 2.5 mA. Find β, α and I_E.

Q4.2 α = 0.99. Find β.

Q4.3 CE amplifier: I_C = 2 mA, R_C = R_L = 4.7 kΩ, emitter bypassed. Find A_v.

Q4.4 CMOS gate: C_L = 10 pF, V_{DD} = 1.8 V, f = 100 MHz, activity α = 0.5. Dynamic power?

Q4.5 NMOS: k = 1 mA/V², V_{th} = 0.7 V, V_{GS} = 2.7 V (saturation). I_D?

Q4.6 Emitter follower voltage gain and why it is useful?

Q4.7 A BJT switch must carry 200 mA, β_{min} = 40, 5 V drive, 3× overdrive. Find R_B.

Q4.8 Which CMOS transistor conducts when the inverter input is HIGH?

5. Op-amps and Regulators

Syllabus: working principle, applications, problem solving – inverting/non-inverting, positive and negative feedback, summer, integrator, differentiator; regulators.

5.1 Ideal op-amp rules

Infinite open-loop gain, infinite input impedance, zero output impedance, infinite bandwidth, infinite CMRR. With **negative feedback**: (1) no current enters the inputs; (2) $V_+ = V_-$ (**virtual short**). With **positive feedback** (or no feedback) the output saturates → comparator / Schmitt trigger / oscillator.

Key formulae	
Inverting amp	$V_o = -(R_f/R_{in})V_{in}$; $Z_{in} = R_{in}$
Non-inverting amp	$V_o = (1 + R_f/R_g)V_{in}$; Z_{in} very high; buffer: gain = 1
Inverting summer	$V_o = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3 \ldots)$
Difference amp	$V_o = (R_2/R_1)(V_2 - V_1)$ with matched resistor pairs
Integrator	$V_o(t) = -(1/RC)\int V_{in}dt$; DC input → ramp of slope $-V_{in}/RC$
Differentiator	$V_o = -RC \cdot dV_{in}/dt$ (amplifies noise → add small series R)
Closed-loop gain (feedback)	$A_f = A/(1 + A\beta) \approx 1/\beta$ for large loop gain; GBW = $A_{CL} \times$ BW (constant)
Slew rate	SR = max dV_o/dt; max undistorted frequency $f_{max} = SR/(2\pi V_p)$
CMRR	$CMRR(dB) = 20 \log_{10}(A_d/A_{cm})$
Inverting Schmitt trigger	$V_{TH} = \pm V_{sat} \cdot R_1/(R_1 + R_2)$ (R_1 to ground, R_2 from output to + input)
Instrumentation amp	$A_v = (1 + 2R/R_{gain}) \cdot (R_3/R_2)$ – high CMRR, high Z_{in}

- **Negative feedback** gives stable, predictable gain, wider bandwidth, lower distortion and controlled Z_{in}/Z_{out}; **positive feedback** gives hysteresis (Schmitt trigger – noise-immune comparator), latching and oscillators (Barkhausen: loop gain = 1, phase = 0°/360°).
- **Comparator** has no feedback or positive feedback and is designed to saturate fast; an op-amp used as a comparator is slower. Open-collector comparators (LM339) need a pull-up.
- Real-world errors: input offset voltage, input bias current, finite GBW, slew rate, output swing limits (rail-to-rail?), noise.

5.2 Voltage regulators

Type	**How it works**	**Pros / cons**
Zener shunt	Zener + series R	Simple; wasteful, poor load regulation
Linear series (78xx, LM317, LDO)	Pass transistor controlled by error amp against a reference	Low noise, simple, fast; **$\eta = V_{out}/V_{in}$**, heat = $(V_{in} - V_{out}) \cdot I_{load}$; needs dropout (78xx ≈ 2 V, LDO ≈ 0.1–0.3 V)
Switching buck (step-down)	Inductor + switch + diode/sync FET, PWM	85–95 % efficient; more noise/EMI, needs layout care
Switching boost / buck-boost	Inductor energy transfer	Step-up or inverting; same trade-offs

Key formulae	
Linear regulator	$P_{diss} = (V_{in} - V_{out})I_L$; $\eta = V_{out}/V_{in}$ (ignoring I_q)
Buck (CCM)	$V_{out} = D \cdot V_{in}$
Boost (CCM)	$V_{out} = V_{in}/(1 - D)$
Buck-boost (inverting)	$V_{out} = -V_{in} \cdot D/(1 - D)$
LM317	$V_{out} = 1.25\ V \times (1 + R_2/R_1) + I_{adj}R_2$ ($R_1 \approx 240\ \Omega$)
Line regulation	$\Delta V_{out}/\Delta V_{in}$; **Load regulation** = $\Delta V_{out}/\Delta I_{load}$ (or %)

Worked example – Op-amp circuits

Problem: (a) Inverting amp R_f = 100 kΩ, R_{in} = 10 kΩ, V_{in} = 0.2 V. (b) Summer, all resistors 10 kΩ, inputs 1 V, 2 V, 3 V. (c) Integrator R = 10 kΩ, C = 0.1 µF with 1 V DC input.

Solution: (a) gain −10 → **−2 V**. (b) $V_o = -(1+2+3) =$ **−6 V**. (c) RC = 1 ms → slope = −1/1 ms = **−1000 V/s** (−1 V every ms) until it saturates at the rail.

Worked example – Slew-rate limit

Problem: SR = 0.5 V/µs. Maximum undistorted frequency for a 5 V peak sine?

Solution: $f_{max} = SR/(2\pi V_p) = 0.5\times10^6/(2\pi\times5)$ = **15.9 kHz**.

Worked example – Regulator choice

Problem: Convert 12 V → 5 V at 500 mA. Compare linear and buck (90 % efficient).

Solution: Linear: $P_{diss} = 7 \times 0.5$ = **3.5 W**, η = 41.7 % (needs a heat sink). Buck: P_{out} = 2.5 W, P_{in} = 2.78 W, loss = **0.28 W**; D = 5/12 = 0.417.

Practice problems (answers in the Answer Key)

Q5.1 Non-inverting amp, R_f = 100 kΩ, R_g = 10 kΩ, V_{in} = 0.5 V. V_o?

Q5.2 Design an inverting amp with gain −5 and Z_{in} = 2 kΩ.

Q5.3 GBW = 1 MHz; closed-loop gain 50. Bandwidth?

Q5.4 SR = 1 V/µs; 10 V_{pp} sine. Highest undistorted f?

Q5.5 Inverting Schmitt, V_{sat} = ±12 V, R_1 = 1 kΩ, R_2 = 9 kΩ. Thresholds?

Q5.6 Buck converter 24 V → 3.3 V. Duty cycle?

Q5.7 Linear regulator 9 V → 5 V at 200 mA. Power lost and efficiency?

Q5.8 A_d = 1000, A_cm = 0.1. CMRR in dB?

Q5.9 Difference amp R_1 = 10 kΩ, R_2 = 100 kΩ, $V_2 - V_1$ = 0.3 V. V_o?

Q5.10 Integrator R = 100 kΩ, C = 1 µF, input +2 V DC. Output slope?

Q5.11 LM317 with R_1 = 240 Ω: choose R_2 for 5 V.

Q5.12 Boost converter 5 V → 12 V. Duty cycle?

6. Filter Circuits – Active and Passive

Syllabus: LPF, HPF, waveforms and working.

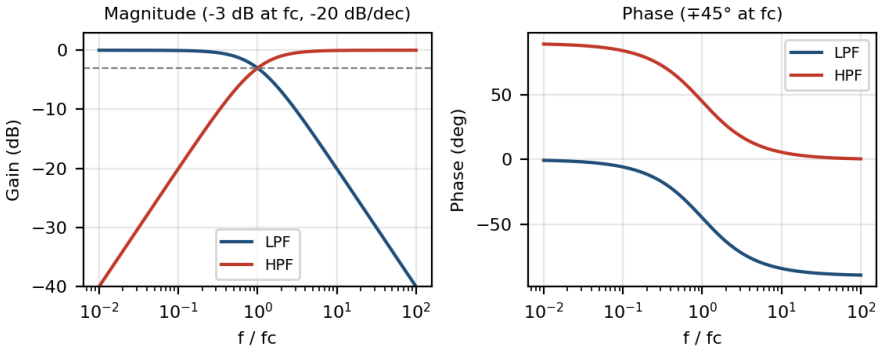


Fig 6.1 – First-order low-pass and high-pass Bode plots.

Key formulae	
RC low-pass	$H(j\omega) = 1/(1 + j\omega RC)$; $\lvert H\rvert = 1/\sqrt{(1 + (f/f_c)^2)}$; $\varphi = -\arctan(f/f_c)$
RC high-pass	$\lvert H\rvert = (f/f_c)/\sqrt{(1 + (f/f_c)^2)}$; $\varphi = +\arctan(f_c/f)$
Cut-off	$f_c = 1/(2\pi RC)$; at f_c gain = 0.707 (−3 dB), phase ∓45°
Roll-off	−20 dB/decade (−6 dB/octave) per order; n-th order: −20n dB/dec
dB	$dB = 20\log_{10}(V_{out}/V_{in}) = 10\log_{10}(P_{out}/P_{in})$; −3 dB ≙ 0.707, −6 dB ≙ 0.5, −20 dB ≙ 0.1
Step response	$t_r(10–90\ \%) = 2.2RC = 0.35/f_c$
Active LPF (1st order, inverting)	$H = -(R_f/R_{in})/(1 + j\omega R_f C_f)$; $f_c = 1/(2\pi R_f C_f)$
Sallen-Key LPF (unity gain, $R_1 = R_2 = R$, C_1 to output, C_2 to ground)	$f_c = 1/(2\pi R\sqrt{(C_1 C_2)})$; $Q = \tfrac{1}{2}\sqrt{(C_1/C_2)}$; Butterworth (Q = 0.707) needs $C_1 = 2C_2$
2nd-order Butterworth	$\lvert H\rvert = 1/\sqrt{(1 + (f/f_c)^4)}$; −40 dB/dec
Band-pass	$f_0 = \sqrt{(f_L f_H)}$; $BW = f_H - f_L$; $Q = f_0/BW$

Waveform behaviour (very common interview question)

Input	**RC low-pass (output across C)**	**RC high-pass (output across R)**
Sine, $f \ll f_c$ / $f \gg f_c$	Passes (gain ≈ 1) / attenuated	Attenuated / passes
Square wave, period ≫ τ	Edges are rounded – exponential charge/discharge; reaches full amplitude	Output is a spike at each edge, decaying to 0 (differentiator)
Square wave, period ≪ τ	Small triangular wave (integrator) around the average	Square wave passes with a droop (tilt); DC removed
DC	Passes	Blocked

	Passive	**Active (op-amp based)**
Components	R, L, C only	R, C + op-amp (no inductors needed)
Gain	≤ 1 (insertion loss)	Can have gain; buffered, load-independent
Power / frequency	No supply; works to very high f and high power	Needs supply; limited by op-amp GBW; low-power
Use	RF, simple anti-alias, power line filter	Audio, sensor conditioning, anti-aliasing before ADC

Worked example – RC filter

Problem: R = 1.59 kΩ, C = 100 nF. Find f_c, gain at 10 kHz and rise time.

Solution: $f_c = 1/(2\pi \times 1590 \times 10^{-7}) \approx$ **1.0 kHz**. At 10 kHz: $\lvert H\rvert = 1/\sqrt{(1+100)} = 0.0995 \rightarrow$ **−20 dB**. $t_r = 2.2RC =$ **350 µs**.

Worked example – Sallen-Key Butterworth
Problem: R = 10 kΩ, C_2 = 10 nF, C_1 = 20 nF. Find f_c and Q.
Solution: $f_c = 1/(2\pi \cdot 10^4 \cdot \sqrt{(2 \times 10^{-16})})$ = **1.125 kHz**; Q = ½√2 = **0.707** (maximally flat).

Practice problems (answers in the Answer Key)

Q6.1 Design an RC high-pass with f_c = 100 Hz using C = 0.1 µF.
Q6.2 First-order LPF: gain at f = 5f_c?
Q6.3 Phase of an RC low-pass at f = f_c/10?
Q6.4 2nd-order Butterworth LPF, f_c = 1 kHz. Gain at 4 kHz?
Q6.5 A filter has −3 dB bandwidth 10 kHz. Estimate rise time.
Q6.6 A 1 kHz, 5 V square wave is applied to an RC low-pass with RC = 10 ms. What is the output?
Q6.7 Why would you put a low-pass filter before an ADC?
Q6.8 Band-pass with f_L = 900 Hz, f_H = 1.1 kHz. Find f_0, BW, Q.

7. Digital ICs – Logic, Sequential Circuits, Buffers, Comparators

Syllabus: logic, programmable, counter, shift register, flip-flop, buffer, driver, comparator – working, difference and application.

7.1 Combinational building blocks

- **Universal gates:** NAND and NOR can build any function. NOT = NAND with inputs tied; AND = 2 NANDs; OR = 3 NANDs; XOR = 4 NANDs.
- **De Morgan:** $(A \cdot B)' = A' + B'$; $(A + B)' = A' \cdot B'$. **Useful identities:** $A + AB = A$; $A + A'B = A + B$; $AB + AB' = A$.
- **Mux** (2^n:1 needs n select lines) – data selector, can implement any n-variable function; **decoder** ($n \to 2^n$), **encoder**; **half adder** $S = A \oplus B$, $C = AB$; **full adder** $S = A \oplus B \oplus C_{in}$, $C_{out} = AB + C_{in}(A \oplus B)$.
- **Logic family terms:** noise margins $NM_H = V_{OH} - V_{IH}$, $NM_L = V_{IL} - V_{OL}$; fan-out = how many inputs one output can drive; power–delay product.

7.2 Latches, flip-flops, counters, shift registers

Element	Characteristic equation / behaviour
SR latch	$Q^+ = S + R'Q$ ($S = R = 1$ forbidden)
D flip-flop	$Q^+ = D$ (captures D at clock edge) – the standard building block of registers
JK flip-flop	$Q^+ = JQ' + K'Q$; $J = K = 1$ toggles; $J = K = 0$ holds
T flip-flop	$Q^+ = T \oplus Q$; $T = 1$ toggles → divide-by-2
Latch vs flip-flop	**Latch** is level-sensitive (transparent while enable is high); **flip-flop** is edge-triggered (samples only at the clock edge)

Counter / register	Facts and applications
Asynchronous (ripple) counter	Each FF clocks the next; simple but delays accumulate (glitches); used for frequency division
Synchronous counter	All FFs share the clock; faster, glitch-free decoding; needs more logic
Mod-N counter	Needs $n = \lceil \log_2 N \rceil$ flip-flops; $f_{out} = f_{clk}/N$; decade counter = mod-10 (4 FFs)
Ring counter	N FFs → N states (one-hot); no decoding needed
Johnson (twisted-ring) counter	N FFs → 2N states; only one bit changes per step (glitch-free decode)
Shift register (SISO, SIPO, PISO, PIPO)	SIPO = serial-to-parallel (e.g. 74HC595 for GPIO expansion), PISO = parallel-to-serial; n clocks to shift n bits; delay line; serial communication, LFSR/CRC/PRBS generation

7.3 Buffer, driver, comparator, programmable logic – differences

Device	What it does	Application
Buffer	Non-inverting (or inverting) stage that regenerates logic levels and isolates source from load; increases fan-out	Clock distribution, bus isolation, level restoration
Line / bus driver (74x244, 74x245, ULN2003)	Buffer with higher current drive (and often tri-state for shared buses)	Drive LEDs, relays, long cables, bus transceiver
Tri-state / open-drain	Tri-state has a third high-Z state; open-drain/collector pulls low only and needs a pull-up (allows wired-AND and level shifting)	Shared buses, I²C
Analog comparator (LM339, LM393)	Compares two analog voltages → digital output	Threshold detect, zero-crossing, ADC front-end, PWM generation
Digital (magnitude) comparator (74x85)	Compares two binary words → A>B, A=B, A<B	Address decode, sorting, thresholding
Programmable logic (PAL, PLA, CPLD, FPGA)	PAL: programmable AND plane, fixed OR; PLA: both programmable; CPLD: non-volatile macrocells (instant-on); FPGA: LUT + flip-flop fabric, SRAM based, reconfigurable	Glue logic, prototyping, accelerators, test fixtures

Worked example – Counters

Problem: (a) A 4-bit binary counter is clocked at 16 MHz – frequency at the MSB? (b) How many FFs for a mod-60 counter? (c) A Johnson counter with 5 FFs has how many states?

Solution: (a) 16 MHz/2^4 = **1 MHz**. (b) $2^6 = 64 \geq 60 \rightarrow$ **6 FFs**. (c) 2 × 5 = **10 states** (decade counter without decoding glitches).

Worked example – Boolean simplification

Problem: Simplify F = AB + AB′ + A′B.

Solution: AB + AB′ = A, so F = A + A′B = **A + B** (OR gate).

Practice problems (answers in the Answer Key)

Q7.1 4-bit ripple counter input 32 MHz. Output frequency at Q3?

Q7.2 Minimum flip-flops to count 100 states?

Q7.3 A 6-FF Johnson counter has how many unique states?

Q7.4 TTL levels: V_{OH} = 2.4, V_{IH} = 2.0, V_{OL} = 0.4, V_{IL} = 0.8. Find noise margins.

Q7.5 Implement OR using only NAND gates. How many?

Q7.6 JK flip-flop with J = K = 1 and clock of 1 kHz. Output frequency?

Q7.7 How many select lines does an 8:1 mux need?

Q7.8 Why must an open-collector output have a pull-up?

Q7.9 Difference between latch and flip-flop?

Q7.10 A 8-bit shift register is clocked at 1 MHz. Time to shift in one byte?

8. Digital IC Characteristics – Timing Parameters

Syllabus: setup/hold time, rise and fall time, bandwidth, propagation delay.

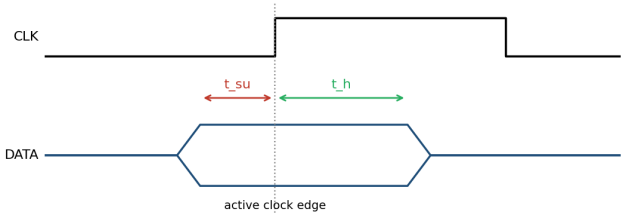


Fig 8.1 – Setup time (data stable before the clock edge) and hold time (stable after it).

Parameter	Definition
Setup time t_{su}	Minimum time the data must be stable **before** the active clock edge
Hold time t_h	Minimum time the data must stay stable **after** the clock edge
Clock-to-Q t_{cq}	Delay from clock edge to a valid output
Propagation delay t_{pd}	Input-to-output delay, measured between 50 % points (t_{pHL}, t_{pLH}; t_{pd} = average)
Rise time t_r / fall time t_f	Time to go between 10 % and 90 % (some specs use 20–80 %) of the swing
Bandwidth	BW ≈ 0.35/t_r (single-pole); knee frequency of a digital edge ≈ 0.5/t_r
Skew / jitter	Skew: static clock arrival difference; jitter: cycle-to-cycle variation of the edge
Pulse width, duty cycle	Duty = $t_{high}/T \times 100$ %; many ICs specify minimum pulse width
Metastability	If setup/hold is violated the FF output may hover mid-level for an unpredictable time → use a 2-flop synchroniser for async inputs

Key formulae	
Max clock frequency (single stage)	$T_{clk} \geq t_{cq} + t_{pd,max}(\text{logic}) + t_{su} + t_{skew}$; $f_{max} = 1/T_{min}$
Setup slack	$\text{Slack} = T_{clk} - (t_{cq} + t_{pd,max} + t_{su} + t_{skew}) \geq 0$
Hold condition	$t_{cq,min} + t_{pd,min} \geq t_h + t_{skew}$ (**independent of clock frequency**)
Rise time / bandwidth	$t_r = 0.35/BW$; $t_r = 2.2RC$
Measured rise time	$t_{r,meas} = \sqrt{(t_{r,signal}^2 + t_{r,scope}^2 + t_{r,probe}^2)}$
Frequency / duty	$f = 1/T$; duty = t_{on}/T
MTBF (synchroniser)	$\text{MTBF} = e^{t_r/\tau} / (T_0 \cdot f_{clk} \cdot f_{data})$ – grows exponentially with resolution time

Worked example – Setup / hold analysis

Problem: t_{cq} = 0.5 ns, $t_{pd,max}$ = 3 ns, t_{su} = 0.3 ns, skew 0.2 ns. Find f_{max}. Hold check: $t_{cq,min}$ = 0.3 ns, $t_{pd,min}$ = 0.2 ns, t_h = 0.4 ns.

Solution: T_{min} = 0.5 + 3 + 0.3 + 0.2 = 4.0 ns → $\mathbf{f_{max} = 250\ MHz}$. Hold: 0.3 + 0.2 = 0.5 ns vs t_h + skew = 0.6 ns → **violation by 0.1 ns**. Fix: insert delay (buffers) on the data path – slowing the clock does NOT fix hold.

Worked example – Scope rise time

Problem: A signal has t_r = 2 ns; the scope has t_r = 3.5 ns (100 MHz). What does it display?

Solution: $\sqrt{(2^2 + 3.5^2)}$ = **4.03 ns**. That is why the scope needs BW ≥ 3–5× the signal's BW.

Practice problems (answers in the Answer Key)

Q8.1 t_{cq} = 0.4, t_{pd} = 2.6, t_{su} = 0.5 ns, no skew. f_{max}?

Q8.2 Same path at 200 MHz. Setup slack?

Q8.3 A signal rise time is 35 ns. Required bandwidth?

Q8.4 200 MHz scope (t_r = 1.75 ns), 1 ns signal edge. Displayed t_r?

Q8.5 Hold check: $t_{cq,min}$ = 0.25, $t_{pd,min}$ = 0.1, t_h = 0.3, skew 0.1 ns. Pass or fail?

Q8.6 A clock is high for 30 ns in a 100 ns period. Duty cycle and frequency?

Q8.7 Why does an asynchronous input need a 2-flop synchroniser?

Q8.8 Does reducing clock frequency fix a hold violation? Why?

9. ADC and DAC – Static and Dynamic Parameters

Syllabus: understanding of ADC/DAC static and dynamic parameters.

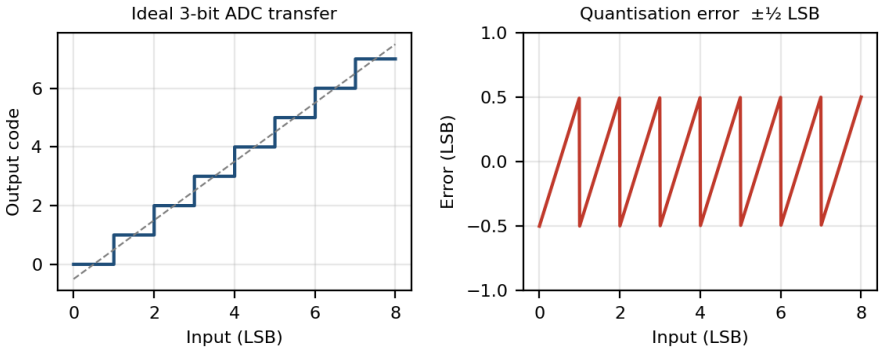


Fig 9.1 – Ideal 3-bit ADC: each code spans 1 LSB; quantisation error is ±½ LSB (sawtooth).

Key formulae	
LSB (step size)	$LSB = V_{FS}/2^N$ ($\approx V_{ref}/2^N$); number of codes = 2^N
ADC code	$D = \text{floor}(V_{in}/LSB)$ (or round, depending on offset convention)
DAC output	$V_{out} = V_{ref} \cdot D/2^N$
Quantisation error	±½ LSB; quantisation noise rms = LSB/√12
Ideal SNR	SNR = 6.02N + 1.76 dB (full-scale sine)
ENOB	ENOB = (SINAD − 1.76)/6.02
Nyquist	$f_s \geq 2 \cdot f_{max}$; anything above $f_s/2$ aliases
DNL / INL	DNL = (actual step width / LSB) − 1; INL = cumulative deviation from ideal straight line; DNL ≤ −1 LSB → missing code; DNL < 1 LSB + monotonic ≙ no missing codes
Jitter-limited SNR	$SNR = -20 \log_{10}(2\pi \cdot f_{in} \cdot t_j)$
THD	$THD = 20 \log(\sqrt{(V_2^2 + V_3^2 + \ldots)}/V_1)$
Oversampling	Each 4× oversampling (plain) ≈ +6 dB ≈ +1 bit; noise shaping (Σ-Δ) does much better
R-2R DAC	$V_{out} = V_{ref} \cdot \Sigma(b_i \cdot 2^i)/2^N$ – only two resistor values, good matching

Parameter	**Static (DC) specs**	**Dynamic (AC) specs**
Examples	Offset error, gain error, INL, DNL, monotonicity, missing codes, zero/full-scale error	SNR, SINAD, THD, SFDR, ENOB, bandwidth, settling time, glitch energy (DAC), aperture jitter, conversion time, sample rate
Measured with	Slow ramp / histogram / DC code transitions	Sine wave input + FFT

ADC type	**Speed / resolution**	**Notes**
Flash (parallel)	Fastest (GSPS); 6–8 bit	$2^N - 1$ comparators – size/power explode with N
SAR	Medium (kSPS–10s MSPS); 8–18 bit	Binary search, N clocks per sample; most popular in MCUs
Pipeline	High speed (10–500 MSPS); 8–16 bit	Multi-stage, latency of several clocks
Σ-Δ (sigma-delta)	Low speed, very high resolution (16–24+ bit)	Oversampling + noise shaping + digital filter; audio, sensors
Dual-slope / integrating	Slow, high accuracy	Good noise rejection – multimeters

Worked example – 12-bit ADC

Problem: V_{ref} = 3.3 V, N = 12. Find LSB, the code for 1.2 V, and the ideal SNR. If the measured SINAD is 68 dB, find ENOB.

Solution: LSB = 3.3/4096 = **0.806 mV**. Code = 1.2/0.000806 = 1489.5 → **1489**. SNR = 6.02×12 + 1.76 = **74 dB**. ENOB = (68 − 1.76)/6.02 = **11.0 bits**.

Worked example – DAC and jitter
Problem: (a) 8-bit DAC, V_{ref} = 5 V, code 200 → V_{out}? (b) 10 MHz input with 1 ps rms clock jitter – SNR limit?
Solution: (a) 5 × 200/256 = **3.906 V**. (b) $-20\log(2\pi \times 10^{7} \times 10^{-12}) = -20\log(6.28\times10^{-5})$ = **84 dB**.

Practice problems (answers in the Answer Key)

Q9.1 10-bit ADC with 5 V reference. LSB and max quantisation error?
Q9.2 Ideal SNR of a 14-bit ADC?
Q9.3 Minimum sampling rate for audio up to 20 kHz?
Q9.4 An ADC step is 1.4 LSB wide. DNL? A zero-width step?
Q9.5 How many comparators in an 8-bit flash ADC?
Q9.6 12-bit SAR ADC at 1 MSPS: minimum internal clock?
Q9.7 8-bit DAC, 2.56 V reference, digital input 0x80. Output?
Q9.8 SINAD of 74 dB. ENOB?
Q9.9 To gain 2 extra bits by plain oversampling, how much must you oversample?
Q9.10 Which ADC architecture suits a digital multimeter, and which a 5 GSPS oscilloscope?

10. Analog vs Digital – Modes, Differences, Advantages

Syllabus: analog and digital modes, differences, advantages.

	Analog	Digital
Signal	Continuous in time and amplitude	Discrete levels (binary 0/1), sampled in time
Noise immunity	Poor – noise adds to the signal permanently	Excellent – noise margin lets receiver regenerate exact bits
Precision	Limited by components, drift, temperature	Set by word length; error-correction possible
Storage / processing	Hard; degrades on copy	Easy, lossless copy, programmable (MCU/DSP/FPGA)
Speed / bandwidth	Can be very fast (RF) with continuous signal	Limited by clock and quantisation
Power / area	Often lower for simple tasks; scales poorly in advanced nodes	Scales well (Moore's law)
Examples	Op-amps, sensors, LDOs, PLLs, RF, ADC/DAC front-ends	Gates, MCU, memories, FPGAs, interfaces

Mixed-signal chips (ADC, DAC, PLL, PMIC, SerDes, sensors) combine both and are where much silicon validation effort goes: analog blocks need characterisation over voltage/temperature, digital blocks need functional and timing validation.

Common digital communication interfaces (frequently asked in validation interviews)

Interface	Wires	Key facts
UART	TX, RX (async)	No clock; both ends agree on baud rate; 8N1 frame = 1 start + 8 data + 1 stop = 10 bits; effective bytes/s = baud/10
SPI	SCLK, MOSI, MISO, CS	Synchronous, full duplex, fast (10s of MHz), master selects slave with CS; modes 0–3 (CPOL, CPHA)
I²C	SDA, SCL (open-drain)	Needs pull-ups; addressable multi-device bus; 100 kHz / 400 kHz / 1 MHz; ACK after each byte; start/stop conditions
GPIO / PWM	single pin	Digital I/O; PWM duty cycle controls average voltage = $D \cdot V_{DD}$

Worked example – Baud rate

Problem: UART at 115200 baud, 8N1. How many bytes per second and how long for 1 KB?

Solution: 10 bits per byte → **11 520 bytes/s**. 1024 bytes → 0.089 s ≈ **89 ms**.

Practice problems (answers in the Answer Key)

Q10.1 A UART runs at 9600 baud, 8N1. Time to send one byte?

Q10.2 Why does digital data survive a noisy long cable better than an analog voltage?

Q10.3 PWM at 3.3 V with 25 % duty. Average voltage?

Q10.4 Why are I²C lines open-drain?

Q10.5 Name one block that must be analog even in an all-digital SoC.

11. C Programming and Logical Thinking

Syllabus: C-programming or other programming skills for logical-thinking evaluation. Silicon-validation work is full of register access, bit manipulation and test scripts, so focus on the topics below.

11.1 Concepts to know

- **Data types:** char (8), short (16), int (usually 32), long, float, double; use `stdint.h` types (uint8_t, uint32_t) for hardware. Signed integer overflow is undefined; unsigned wraps.
- **Storage classes:** `static` (local: retains value between calls; global: file scope), `extern`, `const` (read-only), `volatile` (compiler must not optimise away reads/writes – needed for hardware registers, ISR-shared variables).
- **Pointers:** p+1 moves by sizeof(*p); arr[i] ≡ *(arr+i); pointer to function; NULL; dangling pointers; pointer-to-register: `(*(volatile uint32_t*)0x40021000)`.
- **Bitwise operators:** & | ^ ~ << >>; a left shift by n multiplies by 2^n; right shift of a signed negative value is implementation-defined.
- **struct / union / bit-fields:** struct has padding/alignment; union members share memory (used to inspect bytes); **endianness** = byte order (x86/ARM usually little-endian).
- **Memory:** stack (locals), heap (malloc/free), data/bss (globals), text (code). Stack overflow, memory leak, buffer overflow are classic bugs.
- **Preprocessor:** macros need parentheses – `#define SQR(x) ((x)*(x))`; include guards.

11.2 Must-know bit-manipulation snippets

```
x |=  (1u << n);          // set bit n
x &= ~(1u << n);          // clear bit n
x ^=  (1u << n);          // toggle bit n
bit = (x >> n) & 1u;      // read bit n
field = (x >> pos) & ((1u << width) - 1);        // extract a bit-field
x = (x & ~(mask << pos)) | ((val & mask) << pos); // write a bit-field

int is_pow2(uint32_t x)  { return x && !(x & (x - 1)); }

int count_ones(uint32_t x) {          // Kernighan: loops once per set bit
    int c = 0;
    while (x) { x &= (x - 1); c++; }
    return c;
}

uint8_t reverse_bits(uint8_t b) {
    uint8_t r = 0;
    for (int i = 0; i < 8; i++) { r = (r << 1) | (b & 1); b >>= 1; }
    return r;
}

int is_little_endian(void) { uint16_t v = 1; return *(uint8_t *)&v == 1; }

void swap_xor(int *a, int *b) { if (a != b) { *a ^= *b; *b ^= *a; *a ^= *b; } }

// Memory-mapped register access
#define GPIO_ODR  (*(volatile uint32_t *)0x40020014u)
GPIO_ODR |= (1u << 5);     // drive pin 5 high
```

11.3 Classic logic/algorithm questions

```
// Find the missing number in 1..N  (XOR trick)
int missing(int a[], int n) {           // a has n-1 numbers from 1..n
    int x = 0;
    for (int i = 1; i <= n; i++) x ^= i;
    for (int i = 0; i < n - 1; i++) x ^= a[i];
    return x;
}

// Reverse a string in place
void rev(char *s) {
    char *e = s;  while (*e) e++;  e--;
    while (s < e) { char t = *s; *s++ = *e; *e-- = t; }
}

// Fibonacci (iterative)
```

```
unsigned long fib(int n) { unsigned long a = 0, b = 1;
    while (n--) { unsigned long t = a + b; a = b; b = t; }  return a; }

// Compile-time/Run-time: GCD (Euclid)
int gcd(int a, int b) { return b ? gcd(b, a % b) : a; }

// Reverse a singly linked list
struct node { int v; struct node *next; };
struct node *rev_list(struct node *h) {
    struct node *p = 0;
    while (h) { struct node *n = h->next; h->next = p; p = h; h = n; }
    return p;
}
```

Practice problems (answers in the Answer Key)

Q11.1 What is printed? `unsigned char c = 255; c++; printf("%d", c);`

Q11.2 `int a[5]={1,2,3,4,5}; int *p = a; printf("%d", *(p+2));`

Q11.3 `#define SQR(x) x*x` – value of `SQR(2+3)`?

Q11.4 `char c = -1; unsigned char u = c; printf("%d", u);`

Q11.5 Value of `(uint8_t)(~0x0F)`?

Q11.6 Why is `volatile` needed for a status register polled in a loop?

Q11.7 What does `x & (x-1)` do?

Q11.8 Write an expression to check bit 3 of a register R.

Q11.9 `int x = 5; printf("%d", x++ * 2);` then x?

Q11.10 sizeof of `struct {char a; int b;}` on a 32-bit system?

Q11.11 How do you set bits 7:4 of a register to 0xA without disturbing others?

Q11.12 Give a one-line test for a power of two. Is 0 a power of two by that test?

12. Power Sources, Test and Measurement Equipment

Syllabus: common power source types; test and measurement equipment understanding.

12.1 Power sources

Source	Key facts
Primary battery (alkaline 1.5 V, coin cell Li 3 V)	Not rechargeable; low self-discharge, good for low-power long-life devices
Li-ion / Li-polymer	3.7 V nominal, 4.2 V full, ≈ 3.0 V cut-off; high energy density; needs protection and CC/CV charging
NiMH / NiCd	1.2 V per cell; rechargeable; moderate energy density
Lead-acid	2 V per cell (12 V = 6 cells); cheap, heavy; UPS and automotive
AC mains → adapter	Transformer or SMPS + rectifier; 230 V/50 Hz in India; isolation from mains is a safety requirement
Linear bench supply	Low noise, simple, heavy; excellent for analog work
SMPS / DC-DC converters	High efficiency, compact; switching noise and ripple (check with scope at 20 MHz BW limit)
Programmable bench PSU	CV (constant voltage) and CC (constant current) modes, over-voltage / over-current protection, remote sense, sequencing
USB / PoE / solar	USB 5 V (up to 3 A; PD up to 48 V); PoE over Ethernet; solar needs MPPT

Key formulae	
Battery runtime	$t(h) = \text{capacity (Ah)}/I_{avg}$ (derate 10–30 % for real conditions)
Energy	$E(Wh) = V_{nom} \times Ah$
Efficiency	$\eta = P_{out}/P_{in}$; $P_{in} = P_{out}/\eta$
Ripple	ripple % = $V_{ripple,pp}/V_{DC} \times 100$
Shunt current measurement	$I = V_{shunt}/R_{shunt}$ (burden voltage = $I \cdot R_{shunt}$ – keep small)
Probe attenuation	10× probe: R_{probe} = 9 MΩ + scope 1 MΩ → $V_{scope} = V_{in}/10$; C_{in} drops to ≈ 10–15 pF

12.2 Test and measurement instruments

Instrument	Measures / purpose	Key points to remember
Digital multimeter (DMM)	V, I, R, continuity, diode, capacitance	Voltage: parallel (high Z); current: **in series** (low burden); resistance: power off; 4-wire (Kelvin) for milliohms; watch accuracy ±(% reading + digits)
Oscilloscope	Voltage vs time; rise time, frequency, jitter, ripple	Bandwidth ≥ 3–5× highest signal frequency; sample rate ≥ 2.5–5× BW; 10× probe with compensation adjusted using the square wave; short ground spring to avoid ringing; trigger settings; input 1 MΩ‖≈ 15 pF or 50 Ω
Function / arbitrary waveform generator	Sine, square, ramp, pulse sources	Output impedance 50 Ω – double the voltage into high-Z load
Logic analyzer	Many digital channels, timing + protocol decode (I²C/SPI/UART)	State vs timing mode; sample rate vs signal edge; for digital bus debug
Spectrum analyzer	Amplitude vs frequency	Harmonics, EMI, phase noise; RBW/VBW settings
Bench PSU / SMU / source-measure unit	Source V/I and measure I/V simultaneously	I–V curves, leakage, I_{DD} measurement, 4-wire sense
Electronic load	Programmable sink (CC, CV, CR, CP)	Test power supplies, batteries, load-transient response
LCR meter / VNA	Component values; S-parameters, impedance	RF and signal-integrity characterisation
Protocol analyzer / JTAG / debugger	Bus-level and on-chip debug	Run control, register peek/poke, boundary scan

Instrument	Measures / purpose	Key points to remember
Thermal chamber / hot plate	Temperature control for characterisation	Used with PVT sweeps (−40 °C … 125 °C)

Worked example – Battery life and probing

Problem: (a) A 2500 mAh, 3.7 V Li-ion cell powers a board drawing 80 mA. Runtime and energy? (b) A 10 Ω shunt shows 12 mV. Current?

Solution: (a) 2.5 Ah/0.08 A = **31 h** ideal (≈ 25 h after derating); E = 3.7 × 2.5 = **9.25 Wh**. (b) I = 12 mV/10 Ω = **1.2 mA**.

Practice problems (answers in the Answer Key)

Q12.1 A 1000 mAh battery supplies 50 mA continuously. Ideal runtime?

Q12.2 A buck converter delivers 5 V at 2 A with 90 % efficiency. Input power?

Q12.3 Signal has 2 ns rise time. Minimum scope bandwidth (rule of thumb $0.35/t_r$ × safety)?

Q12.4 Why measure current with a series shunt instead of just a DMM sometimes?

Q12.5 A 10× probe reads 200 mV on the scope screen (scope set to 1×). Actual signal?

Q12.6 Which instrument would you use to see an I^2C transaction? A power-supply noise spike?

Q12.7 A function generator set to 1 V into 50 Ω shows what in a high-Z scope input?

13. Silicon Validation Context – What Interviewers Also Ask

Silicon (post-silicon) validation checks that the fabricated chip meets its specification across **PVT – Process, Voltage, Temperature** corners, using lab equipment and software, then debugs failures to root cause.

Term	Meaning
Verification vs validation	**Verification**: pre-silicon, simulation/emulation of RTL against spec. **Validation**: post-silicon, real chip on a board in the lab
Bring-up	First power-on: check power rails, clocks/PLL lock, reset, JTAG access, boot, then basic register read/write
Characterisation	Measure parameters (V_{OH}/V_{OL}, I_{DD}, F_{max}, accuracy, noise, INL/DNL) across PVT and compare with datasheet min/typ/max
PVT corners	Process: SS/TT/FF (slow/typical/fast) lots; Voltage: V_{nom} ±10 %; Temperature: −40, 25, 85, 125 °C
Shmoo plot	2-D pass/fail map vs two parameters (e.g. frequency vs voltage) showing operating margin
Margining	Deliberately stress V, f, T, or timing to measure how much margin exists
DFT / Scan / JTAG / BIST	Design-for-test: scan chains, boundary scan (JTAG), built-in self-test – used for debug and manufacturing test
ATE vs bench	ATE = automated production tester (fast, many devices); bench = flexible manual/automated lab setup for debug
Debug flow	Reproduce → isolate (voltage/temp/frequency dependence, board vs silicon, one unit vs many) → hypothesis → measure → root cause → feedback to design/test
Silicon errata	Documented, confirmed chip bugs and workarounds

Typical interview questions – short model answers

- **The chip works at 25 °C but fails at 125 °C. What do you check?** Slower transistors and higher leakage → timing margin and I_{DD}; check setup paths, reference accuracy, clock/PLL range, supply droop; sweep V and f to find the failing region; compare multiple units.
- **A register read returns wrong data intermittently.** Check clock/reset, signal integrity on the bus (scope, ringing, crosstalk), setup/hold at the receiver, pull-ups, supply noise, address alignment, and compare with a different board.
- **How do you measure I_{DD} of a chip in sleep mode (µA)?** Use an SMU/precision DMM or a low-ohm shunt with differential amplifier; avoid leakage via I/Os and probes; wait for steady state; average.
- **How do you validate an ADC?** Static: slow ramp or code-density histogram → offset, gain, INL, DNL. Dynamic: pure sine at several frequencies → FFT → SNR, THD, SFDR, ENOB. Repeat across PVT.
- **Why use a 50 Ω terminated cable for fast signals?** Prevent reflections: matches the transmission-line impedance.
- **What is a pull-up / pull-down resistor?** Defines the logic level of a floating or open-drain pin.
- **What causes ground bounce / supply droop?** L·di/dt of simultaneous switching through package/PCB inductance – use decoupling capacitors close to the pins.

14. One-Page Cheat Sheet

Topic	Essentials
R, L, C	$\tau_{RC} = RC$, $\tau_{RL} = L/R$; $f_c = 1/(2\pi RC)$; $X_C = 1/(2\pi fC)$; $X_L = 2\pi fL$; $f_0 = 1/(2\pi\sqrt{LC})$; $E_C = \frac{1}{2}CV^2$, $E_L = \frac{1}{2}LI^2$
Network	$V_{div} = V \cdot R_2/(R_1+R_2)$; $I_1 = I \cdot R_2/(R_1+R_2)$; $P_{max} = V_{th}^2/4R_{th}$
Diode	$V_F \approx 0.7$ V (Si); HW: $V_{dc} = V_m/\pi$; FW: $2V_m/\pi$; ripple $V_r = I/(fC)$; $R_s = (V_{in} - V_Z)/(I_Z + I_L)$
BJT	$I_C = \beta I_B$; $I_E = (\beta+1)I_B$; $\alpha = \beta/(\beta+1)$; $g_m = I_C/V_T$; $A_v(CE) = -g_m R_C$; CC gain ≈ 1
MOSFET	$I_D = \frac{1}{2}k(V_{GS} - V_{th})^2$ (sat); $g_m = 2I_D/V_{ov}$; $P_{dyn} = \alpha CV^2 f$
Op-amp	Inv: $-R_f/R_{in}$; Non-inv: $1 + R_f/R_g$; Integrator $-1/(RC)\int$; GBW const; $f_{max} = SR/(2\pi V_p)$; CMRR = 20 $\log(A_d/A_{cm})$
Regulators	Linear: $P = (V_{in} - V_{out})I$; buck $V_{out} = DV_{in}$; boost $V_{in}/(1-D)$; LM317: $1.25(1 + R_2/R_1)$
Filters	-3 dB at f_c; -20 dB/dec per pole; $t_r = 0.35/BW = 2.2RC$; Sallen-Key $f_c = 1/(2\pi R\sqrt{(C_1C_2)})$
Digital	$n = \lceil \log_2 N \rceil$ FFs; Johnson = 2N states; ring = N; D: $Q^+ = D$; T: $Q^+ = T \oplus Q$; JK: $Q^+ = JQ' + K'Q$
Timing	$T \geq t_{cq} + t_{pd} + t_{su} + \text{skew}$; hold: $t_{cq,min} + t_{pd,min} \geq t_h + \text{skew}$
ADC/DAC	LSB = $V_{ref}/2^N$; SNR = 6.02N + 1.76; ENOB = (SINAD − 1.76)/6.02; $f_s \geq 2f_{max}$; DNL = step/LSB − 1
Scope / power	BW ≥ 5×f; $t_r = 0.35/BW$; $t_{meas} = \sqrt{(t_s^2 + t_{scope}^2)}$; runtime = Ah/I; $\eta = P_{out}/P_{in}$
dB	20 log (V ratio) = 10 log (P ratio); −3 dB = 0.707; −6 dB = 0.5; 20 dB = ×10; 40 dB = ×100

Answer Key – Practice Problems

Chapter 1 – R, C, L

Q1.1 $\tau = 1$ ms; $v = 5(1 - e^{-2}) = 5 \times 0.8647 =$ **4.32 V**.

Q1.2 $v = L \cdot \Delta I / \Delta t = 0.1 \times 2/0.04 =$ **5 V**.

Q1.3 $\frac{1}{2}CV^2 = 0.5 \times 10{\times}10^{-6} \times 144 =$ **720 µJ**.

Q1.4 $f_0 = 1/(2\pi\sqrt{(10^{-9})}) =$ **5.03 kHz**; $Q = (1/10)\sqrt{(10^{-3}/10^{-6})} = \sqrt{1000}/10 =$ **3.16**.

Q1.5 $1/(2\pi \cdot 10^4 \cdot 10^{-7}) =$ **159.2 Ω**.

Q1.6 $\tau = L/R =$ **100 µs**.

Q1.7 $(10{\times}40)/(10{+}40) =$ **8 µF**.

Q1.8 Current **leads** voltage by **90°**.

Chapter 2 – Network theory

Q2.1 $5 \times 3.3/5.5 =$ **3.0 V**.

Q2.2 $R = 3/0.01 =$ **300 Ω** (use 330 Ω, giving ≈ 9.1 mA).

Q2.3 $4 \| 12 = 3\ \Omega \rightarrow$ **8 Ω**.

Q2.4 $V_{th} = 6$ V, $R_{th} = 4\text{k} \| 6\text{k} = 2.4$ kΩ; $P_{max} = 36/(4{\times}2400) =$ **3.75 mW**.

Q2.5 KCL: $5 = 2 + 1 + I \rightarrow$ **I = 2 mA**.

Q2.6 $P = 25/220 = 113$ mW < 250 mW → **yes**.

Q2.7 $30/3 =$ **10 Ω** each.

Chapter 3 – Diodes

Q3.1 e^(60/25.85) = e^2.32 ≈ **10×** (one decade per ≈ 60 mV).

Q3.2 $V_{dc} = 10/\pi =$ **3.18 V**; $V_{rms} = 10/2 =$ **5 V**.

Q3.3 $V_{dc} = 20/\pi =$ **6.37 V**; $V_{rms} = 10/\sqrt{2} =$ **7.07 V**.

Q3.4 $I = 3.9/220 =$ **17.7 mA**; $P = 5.1 \times 17.7$ mA = **90 mW**.

Q3.5 **−10 V to 0 V** (DC shift of −5 V, i.e. $-V_m$).

Q3.6 $(5 - 0.7)/1\text{k} =$ **4.3 mA**.

Q3.7 $\mathbf{V_m}$ (centre-tap type needs $2V_m$).

Q3.8 When the transistor turns off, L·di/dt produces a large reverse voltage spike that can destroy the switch; the diode gives the current a safe path.

Chapter 4 – Transistors

Q4.1 $\beta =$ **125**; $\alpha = 125/126 =$ **0.992**; $I_E =$ **2.52 mA**.

Q4.2 $\beta = 0.99/0.01 =$ **99**.

Q4.3 $g_m = 77.4$ mS; $R_C \| R_L = 2.35$ kΩ → $A_v =$ **−182**.

Q4.4 $0.5 \times 10\text{p} \times 3.24 \times 10^8 =$ **1.62 mW**.

Q4.5 $\frac{1}{2} \times 1\text{m} \times 2^2 =$ **2 mA**.

Q4.6 $A_v \approx 1$; high Z_{in}, low Z_{out} → **buffer** between a high-impedance source and a low-impedance load.

Q4.7 $I_B = 5$ mA × 3 = 15 mA; $R_B = 4.3/15\text{m} \approx$ **287 Ω** → 270 Ω.

Q4.8 The **nMOS** (output pulled to GND); pMOS is off.

Chapter 5 – Op-amps & regulators

Q5.1 gain 11 → **5.5 V**.

Q5.2 $R_{in} = 2$ kΩ, $R_f =$ **10 kΩ**.

Q5.3 1 MHz/50 = **20 kHz**.

Q5.4 $V_p = 5$ V: $10^6/(2\pi{\times}5) =$ **31.8 kHz**.

Q5.5 $\pm 12 \times 1/10 =$ **±1.2 V**; hysteresis 2.4 V.

Q5.6 $D = 3.3/24 =$ **0.1375**.

Q5.7 $P = 4{\times}0.2 =$ **0.8 W**; $\eta = 5/9 =$ **55.6 %**.

Q5.8 $20 \log(10^4) =$ **80 dB**.

Q5.9 gain 10 → **3 V**.

Q5.10 −2/(0.1 s) = **−20 V/s** (−20 mV/ms).
Q5.11 $1 + R_2/240 = 4 \rightarrow R_2 =$ **720 Ω**.
Q5.12 D = 1 − 5/12 = **0.583**.

Chapter 6 – Filters

Q6.1 $R = 1/(2\pi \cdot 100 \cdot 10^{-7}) =$ **15.9 kΩ**.
Q6.2 $1/\sqrt{26} = 0.196 \rightarrow$ **−14.2 dB**.
Q6.3 −arctan(0.1) = **−5.7°**.
Q6.4 $1/\sqrt{(1+4^4)} = 0.0624 \rightarrow$ **−24.1 dB**.
Q6.5 0.35/10 kHz = **35 µs**.
Q6.6 τ ≫ period → heavily integrated: small ≈ triangular ripple around 2.5 V (DC average).
Q6.7 To remove frequencies above $f_s/2$ that would otherwise **alias** into the band (anti-aliasing).
Q6.8 $f_0 = \sqrt{(990000)} \approx 995$ Hz; BW = 200 Hz; Q ≈ **4.97**.

Chapter 7 – Digital ICs

Q7.1 32/16 = **2 MHz**.
Q7.2 $2^7 = 128 \geq 100 \rightarrow$ **7**.
Q7.3 **12**.
Q7.4 $NM_H = 0.4$ V; $NM_L = 0.4$ V.
Q7.5 **3**: NOT A, NOT B, then NAND of the two = A + B.
Q7.6 Toggles each clock → **500 Hz** (÷2).
Q7.7 **3**.
Q7.8 It can only pull the line LOW; the resistor provides the HIGH level.
Q7.9 Latch = level-triggered (transparent); flip-flop = edge-triggered.
Q7.10 8 clocks × 1 µs = **8 µs**.

Chapter 8 – Timing

Q8.1 3.5 ns → **285.7 MHz**.
Q8.2 5 − 3.5 = **+1.5 ns**.
Q8.3 0.35/35 ns = **10 MHz**.
Q8.4 $\sqrt{(1 + 3.0625)} =$ **2.02 ns**.
Q8.5 0.35 vs 0.40 → **fails by 0.05 ns**; add ≥ 0.05 ns delay.
Q8.6 **30 %**, **10 MHz**.
Q8.7 The first FF may go metastable; the second gives extra time for it to resolve before the logic uses it.
Q8.8 **No** – hold is checked at the same clock edge; it depends on data-path delay, not period.

Chapter 9 – ADC / DAC

Q9.1 LSB = 4.883 mV; error ±**2.44 mV**.
Q9.2 6.02×14 + 1.76 = **86.04 dB**.
Q9.3 **> 40 kHz** (CD uses 44.1 kHz).
Q9.4 DNL = **+0.4 LSB**; zero-width → DNL = **−1 LSB** (missing code).
Q9.5 $2^8 - 1 =$ **255**.
Q9.6 ≥ 12 cycles per conversion → ≥ **12 MHz** (more once sampling time is included).
Q9.7 128/256 × 2.56 = **1.28 V**.
Q9.8 (74 − 1.76)/6.02 = **12 bits**.
Q9.9 $4^2 =$ **16×**.
Q9.10 Dual-slope / Σ-Δ for the DMM; flash / interleaved pipeline for the scope.

Chapter 10 – Analog vs digital

Q10.1 10/9600 = **1.04 ms**.
Q10.2 The receiver only decides 0/1 using thresholds, so noise smaller than the noise margin is removed completely; analog noise stays in the signal.
Q10.3 **0.825 V**.

Q10.4 Many devices share the line; open-drain + pull-up avoids contention (wired-AND) and allows clock stretching.

Q10.5 PLL/clock generator, ADC/DAC, I/O pads, power management / regulators, sensors.

Chapter 11 – C programming

Q11.1 **0** – unsigned 8-bit wraps around.

Q11.2 **3**.

Q11.3 2+3*2+3 = **11** (not 25) – macros need parentheses.

Q11.4 **255** (bit pattern 0xFF).

Q11.5 **0xF0** (240).

Q11.6 Without it the compiler may cache the value in a CPU register and never re-read the hardware → infinite loop.

Q11.7 Clears the lowest set bit (used to count ones, test power of 2).

Q11.8 `(R >> 3) & 1` or `R & (1u << 3)`.

Q11.9 Prints **10**; x becomes **6** (post-increment).

Q11.10 **8** bytes (3 bytes padding after a for alignment).

Q11.11 `R = (R & ~(0xFu<<4)) | (0xAu<<4);`

Q11.12 `x && !(x & (x-1))`; the `x &&` guard makes 0 return false.

Chapter 12 – Power & T&M;

Q12.1 **20 h**.

Q12.2 10 W/0.9 = **11.1 W** (loss 1.1 W).

Q12.3 Signal BW ≈ 175 MHz → use a **≥ 500 MHz** scope (3–5× factor).

Q12.4 A DMM has burden voltage and range limits; a small shunt + scope/DAQ lets you see dynamic current.

Q12.5 The scope's probe setting must be 10× → actual **2 V**.

Q12.6 Logic analyzer (protocol decode); oscilloscope (short ground lead, 20 MHz BW limit off to see spikes).

Q12.7 **2 V** (the generator expects a 50 Ω load).